Earth Planet. Sci. Lett. 678 (2026) 119816

Contents lists available at ScienceDirect

Earth and Planetary Science Letters

journal homepage: www.elsevier.com/locate/epsl

Discriminating dynamic rupture arrest in fluid-induced microearthquakes using spectral inversion

Francesco Mosconi [a,*], Elisa Tinti [a,b], Mariano Supino [b], Alice-Agnes Gabriel [c,d], Emanuele Casarotti [b], Men-Andrin Meier [e], Antonio Pio Rinaldi [f], Domenico Giardini [e], Massimo Cocco [b]

[a] La Sapienza Università di Roma, Rome, Italy
[b] Istituto Nazionale di Geofisica e Vulcanologia, Rome, Italy
[c] Institute of Geophysics and Planetary Physics, Scripps Institution of Oceanography, University of California, San Diego, California, U.S.A
[d] Department of Earth and Environmental Sciences, Geophysics, Ludwig-Maximilians-Universität Münche, Munich, Germany
[e] Department of Earth and Planetary Sciences, ETH Zurich, Zurich, Switzerland
[f] Swiss Seismological Service (SED), ETH Zurich, Zurich, Switzerland

ARTICLE INFO

Editor: J.P. Avouac

Keywords:
Dynamic rupture
Spectral inversion
Rupture arrest
Induced earthquakes

ABSTRACT

Determining the maximum possible magnitude of fluid-induced earthquakes requires to understand rupture arrest within or outside a fluid-pressurized patch. Recent studies have highlighted the importance of incorporating rupture physics into the study of injection-induced earthquakes. We perform 3D dynamic simulations of spontaneous ruptures propagating across a pressurized fault, stimulated by fluid injection within the nucleation zone. Our simulations unveil two end-member models describing a fluid-induced micro-earthquake: a self-arresting rupture that decelerates spontaneously and a run-away rupture that terminates abruptly at the fault edge. We compute synthetic waveforms radiated from both models and invert them using a probabilistic spectral inversion approach to identify characteristics that distinguish between the two rupture types. We find that self-arresting ruptures radiate less high-frequency waves (with $\gamma > 3$) and lack the typical P/S corner frequency shift. In contrast, run-away ruptures conform to the ω^2 model ($\gamma \sim 2$, $f_c^P / f_c^S \sim 1.3$). We interpret these differences as primarily arising from the rupture arrest mechanism, smooth arrest results in gradual variations in the moment-rate function, whereas abrupt arrest at a barrier causes a sharp changes in the moment-rate function. This abrupt arrest generates high-frequency radiation and a back-propagating stopping phase, playing a critical role in controlling the rupture duration and the radiated seismic waves. Our results demonstrate that spectral features such as high-frequency decay and P/S corner frequency shift may provide observational diagnostics to distinguish rupture arrest mechanisms, even in the absence of direct evidence from rupture kinematics.

1. Introduction

Earthquake ruptures are commonly described as shear dislocations along pre-existing faults, driven by tectonic stresses acting on the surrounding volume and localizing strain on the principal slipping surface (Kanamori and Brodsky, 2004). Dynamic rupture models (Day, 1982) provide a framework to understand how earthquakes nucleate, propagate, and ultimately arrest. While most research has focused on rupture initiation and propagation, rupture arrest remains a less explored mechanism, with notable recent exceptions (e.g., Ke et al., 2022). Rupture ceases when the available energy is no longer sufficient to sustain propagation; however, it remains debated whether arrest is primarily controlled by variations in fault strength, initial stress heterogeneity, or geometric complexities (Das and Aki, 1977; Kame and Yamashita, 1999).

A growing concern in seismic hazard studies is fluid-induced seismicity, which has increased interest in fault mechanics, particularly in the context of micro-earthquakes (Ellsworth, 2013; Lengliné et al., 2014). A fundamental question, central to earthquake physics, is why some ruptures remain confined while others grow into larger events (McGarr, 2014; Galis et al., 2017; Norbeck and Horne, 2018). Addressing this question requires a deeper understanding on how an earthquake stop.

* Corresponding author.
E-mail addresses: francesco.mosconi@uniroma1.it (F. Mosconi), elisa.tinti@uniroma1.it (E. Tinti), mariano.supino@ingv.it (M. Supino), algabriel@ucsd.edu (A.-A. Gabriel), emanuele.casarotti@ingv.it (E. Casarotti), menandrin.meier@eaps.ethz.ch (M.-A. Meier), antoniopio.rinaldi@sed.ethz.ch (A.P. Rinaldi), domenico.giardini@eaps.ethz.ch (D. Giardini), massimo.cocco@ingv.it (M. Cocco).

https://doi.org/10.1016/j.epsl.2025.119816
Received 7 July 2025; Received in revised form 11 December 2025; Accepted 30 December 2025
Available online 16 January 2026
0012-821X/© 2026 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Recently, self-arresting ruptures –events that accelerate beyond the nucleation zone but then decelerate and spontaneously stop over a finite fault area– have been identified in both natural (Xu et al., 2023) and fluid-induced earthquakes (Galis et al., 2017; Maurer et al., 2020; Mosconi et al., 2025). Complementarily, the conditions under which dynamic slip instabilities can nucleate, arrest, and even re-nucleate on a pressurized fault have been elucidated through quasi-static models (Garagash and Germanovich, 2012; Sáez and Lecampion, 2024). These self-arresting ruptures contrast with runaway ruptures, which continue to accelerate until they encounter abrupt variations in stress, strength, or geometry that act as barriers to rupture propagation. 3D numerical simulations of dynamic ruptures nucleating and propagating on a fluid-pressurized patch have successfully reproduced both self-arresting and run-away ruptures, highlighting the role of the spatial stress profile induced by pore pressure changes in controlling rupture arrest (Mosconi et al., 2025). In self-arresting cases, a sufficiently large stress gradient causes gradual deceleration and spontaneous arrest. Distinguishing between run-away and self-arresting ruptures in real earthquakes remains an open challenge, yet it is crucial for seismic hazard assessment, as it directly influences estimates of the maximum expected earthquake magnitude (Maurer et al., 2020). Fracture mechanics-based criteria have been proposed to predict the maximum rupture extent in such scenarios, accounting for the energy balance and stress conditions required for spontaneous arrest (Galis et al., 2017). However, current seismic observations provide limited constraints on rupture arrest characteristics, whether the prevailing mechanism is a gradual energy depletion or an abrupt arrest of the rupture front against an unbreakable barrier. This determines the need for further investigation.

Deepening our understanding of micro-earthquake mechanics is becoming increasingly important, both to constrain the scaling laws that govern seismic source parameters (e.g., Gabriel et al., 2024) and to assess anthropogenic hazard, with the goal of ensuring the safe exploitation of geo-resources. In this context, fluid-induced seismicity, with magnitudes ranging from $M \sim -4$ to 2, offers an unprecedented opportunity to study rupture processes in controlled environments such as geothermal reservoirs and hydraulic fracturing operations. Recent advances in seismic monitoring have significantly improved our ability to analyze micro-earthquakes. High-resolution seismic arrays, particularly those deployed in boreholes and near-fault observatories (NFOs), now provide high-quality recordings of thousands of events (Zoback et al., 2011; Chiaraluce et al., 2014; Guglielmi et al., 2015). A prime example is the Bedretto Underground Laboratory for Geosciences and Geoenergies (BULGG) in the Swiss Alps (Plenkers et al., 2023). Located at depths of 1000–1500 m, BULGG hosts the FEAR ERC-Synergy project, which aims to induce $M_w \sim 1$ earthquakes by reactivating a natural fault under controlled conditions. These events will be recorded using an extensive, multi-disciplinary, on-fault monitoring system, providing a unique dataset to investigate rupture dynamics (Achtziger-Zupančič et al., 2024; Volpe et al., 2023).

To this end, spectral analysis of recorded waveforms is widely used to infer earthquake source parameters in the far-field approximation (e.g., seismic moment, corner frequency, rupture dimension, and stress drop) and investigate the scaling of stress drop with earthquake size (Abercrombie, 1995; Prieto et al., 2004; Abercrombie, 2021). Stress drop estimates for individual seismic sequences and in a limited magnitude range could potentially suggest a dependence on earthquake size (Malagnini et al., 2014). However, early studies (Aki, 1967) and recent literature (Allmann and Shearer, 2009; Cocco et al., 2016; Abercrombie et al., 2025) corroborate a moment-independent scaling for earthquake stress drop across a broad range of magnitudes and with a relatively large variability. Large uncertainties in stress drop estimates complicate interpretations, yet their implications for rupture dynamics and energy dissipation are profound (Abercrombie, 2021). Given that dissipated energy during dynamic breakdown varies with coseismic slip over 13 orders of magnitude in earthquake size (Cocco et al., 2023), while stress drop typically ranges from 0.1 to 100 ~ MPa (Abercrombie, 2021), a better understanding of these scaling features is critical.

In this study, we analyze the seismic radiation from 3D dynamic simulations of self-arresting and run-away ruptures characterizing fluid-induced micro-earthquakes. Our aim is to investigate the source properties of spontaneously decelerating dynamic ruptures through spectral analysis of the recorded waveforms to look for distinctive features compared to the classic crack-like ruptures.

2. 3D Dynamic rupture source models

We conduct 3D simulations to model the dynamic rupture of a pressurized fault patch induced by fluid injection, following the same conditions as in Mosconi et al. (2025), but employing a larger domain, higher spatial resolution, and additional locations for receivers. The pore pressure profile along the fault (axis symmetrical) has been computed through poro-elastic simulation of the fault zone response to fluid injection. The spatially variable pore pressure profile within the pressurized patch induces a gradient in the effective normal stress ($\sigma_n' = \sigma_n - p_f$) across the fault radius. The dynamic rupture simulations are conducted using the open-source ADER-DG (Arbitrary High-Order Derivative Discontinuous Galerkin) software SeisSol (Gabriel et al., 2025); further details are provided in the Supplementary Material. The rupture behavior is governed by a linear slip-weakening constitutive law (Ida, 1972; Andrews, 1976; Day, 1982) (Fig. 1b). In the following, we present two rupture models exhibiting self-arresting and run-away characteristics, which we consider as end-members scenarios within our dynamic rupture modeling framework. Following Mosconi et al. (2025), we select two values of the dynamic friction coefficient ($\mu_d = 0.15$ and $\mu_d = 0.21$) to produce self-arresting and runaway ruptures, respectively. We note that while variations in μ_d effectively control rupture behavior, they are not the only governing factor. Similar dynamic changes can be achieved by modifying other parameters. For example, reducing the critical slip distance D_c can trigger a transition from self-arresting to run-away rupture even with $\mu_d = 0.21$. Likewise, variations in fluid injection properties, such as pressurization level Δp, injection duration, or fault permeability, can alter rupture behavior by changing the absolute stress conditions or pore-pressure gradient.

To analyze seismic radiation from these rupture models, we perform spectral inversion under the point-source assumption using a single-event, multiple-station approach (Supino et al., 2019). Receivers are placed at take-off angles of 90°, 120°, 150°, and 180°, distributed to fully sample all azimuthal angles ϕ around the fault (Fig. 1a).

2.1. Fault model and input parameters

The fault geometry and stress conditions are designed to replicate a fault zone subjected to the conditions found at the Bedretto Laboratory (Achtziger-Zupančič et al., 2024). Our model simulates fluid injection on a 60° dipping normal fault embedded in a homogeneous 3D elastic medium (Fig. 1a). The compressional (α) and shear wave (β) velocities are 2612 m/s and 1513 m/s, respectively, with a density (ρ) of 2620 kg/m^3. The fault plane measures 50×50 m within a larger $1000 \times 1000 \times 1000$ m computational domain (details about the unstructured tetrahedral mesh are provided in the Supplementary Material).

Before injection, the normal stress (σ_n) is 22.7 MPa, while the initial shear stress (τ_0) is 4.7 MPa. The pore-pressure distribution is derived from coupled hydromechanical TOUGH3-FLAC3D simulations (Rinaldi et al., 2022) and is applied as a static field in the dynamic rupture simulations. Our fluid-injection configuration results in a maximum pore pressure, and hence minimum effective normal stress (σ_n') at the injection point, which then increases radially with distance due to the decreasing Δp, as shown in Fig. 1c. For details on the injection protocol and pore-pressure distribution for dynamic rupture simulations, refer to the Supplementary Material.

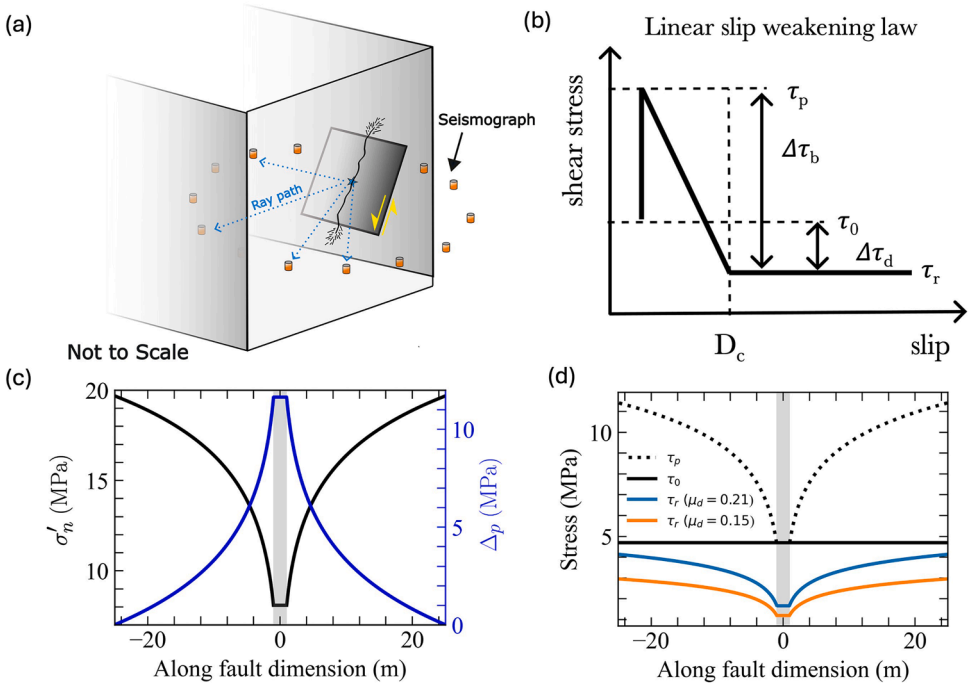


Fig. 1. 3D dynamic rupture model setup. (a) Schematic illustration of the fault geometry and receivers (seismographs) configuration with 90° take-off angle and variable azimuth. (b) Representation of the linear slip weakening constitutive law. (c) Profile of pore pressure change within the 25 m radius pressurized fault patch (blue line), and the corresponding on-fault effective normal stress (black line). The gray bar indicates the location and dimension of the injection borehole. (d) Spatial distribution of the on fault conditions resulting from fluid pressurization. The peak strength (or static fault strength, black dashed line) and the initial shear stress (black solid line) are the same for both self-arresting and run-away rupture models. However, the residual shear stress τ_r differs due to the use of different dynamic friction coefficients (blue solid line: 0.21, orange solid line: 0.15).

The constitutive parameters of the linear-slip-weakening law (Fig. 1b) are: the critical slip distance $D_c = 0.6$ mm, $\mu_s = 0.58$, and variable $\mu_d = 0.21$ or 0.15 to simulate self-arresting and run-away ruptures. Rupture nucleation occurs where the initial shear stress (τ_0) reaches the peak strength ($\tau_p = \mu_s \sigma_n'$) as a result of the reduction in effective normal stress (σ_n'), within a 1 m patch marked by the gray band in Fig. 1c-d. The general on-fault stress conditions are illustrated in Fig. 1d, with further details provided in Mosconi et al. (2025). Here, we simply note that, under this fault parametrization, the maximum residual shear stress ($\tau_r = \mu_d \sigma_n'$) is ~ 4 MPa for $\mu_d = 0.21$ and ~ 3 MPa for $\mu_d = 0.15$. Since $\tau_0 > \tau_r$, the system operates in an unstable regime, allowing for both self-arresting and runaway ruptures (Garagash and Germanovich, 2012; Sáez and Lecampion, 2024).

2.2. Self-arresting rupture

The self-arresting dynamic rupture model corresponds to an $M_w = 0.76$ event characterized by the fault slip distribution shown in Fig. 2a, with an approximately circular shape and a total rupture radius (r) of ≈ 18 m. The peak slip value is ≈ 6 mm at the hypocenter at the fault center. The evolution of the slip velocity ($\dot{\delta}$) for this self-arresting rupture is illustrated in Fig. 2c. This figure shows the temporal evolution of $\dot{\delta}$ along the down-dip direction, highlighting an initial crack expansion accompanied by an increase in both rupture speed v_r and $\dot{\delta}$ within the first ≈ 6 m of the crack length. This is followed by a significant deceleration phase, where the peak slip velocity $\dot{\delta}_{peak}$ within the cohesive zone decreases spatially along with v_r. Notably, Fig. 2c also reveals that, during the deceleration phase, the $\dot{\delta}$ at the crack's center (i.e., the residual slip rate) gradually approaches zero (that is, the rupture approaches its arrest, resembling a pulse-like rupture, see Figure S1 in Supplementary Materials).

In this scenario, the arrest of dynamic rupture occurs smoothly within the pressurized patch and nearly simultaneously at different radial directions. This gradual arrest is governed by the spatial increase of the strength parameter $S = (\tau_p - \tau_0)/(\tau_0 - \tau_r)$, caused by the peak and residual stress gradients (Fig. 1d) induced by pore-pressure perturbations. Specifically, the propagating rupture front experiences an increase in the breakdown stress drop ($\Delta\tau_b = \tau_p - \tau_r$) and a simultaneous reduction in the dynamic stress drop ($\Delta\tau_d = \tau_0 - \tau_r$). This interplay leads to a spatial decrease in the non-dimensional parameter: $R = 1/(1+S) = \Delta\tau_d/\Delta\tau_b$ (Aochi and Madariaga, 2003), ultimately causing the reduction in v_r and the arrest. The final outcome is a rupture that (with this parametrization) does not propagate outside the pressurized fault patch, spontaneously arrests over an extended fault area, thereby maintaining a smaller magnitude. The modeling of a potential re-nucleation of a self-arresting rupture, driven by the temporal evolution of pore pressure and quasi-static slip (Garagash and Germanovich, 2012), is beyond the scope of the present study, as such processes could eventually trigger a subsequent dynamic instability implying a composite rupture behavior.

2.3. Run-away rupture

The run-away dynamic rupture model represents a fluid-induced rupture that does not exhibit spontaneous arrest within the pressurized patch. This rupture event produces a final $M_w = 1.32$, rupturing the entire fault plane with a peak slip of ≈ 12 mm at its center, as shown in the slip distribution of Fig. 2b.

Fig. 2d depicts the time evolution of $\dot{\delta}$ in the down-dip direction, showing a continuous rupture propagation after the first rapid acceleration, along with an increase in $\dot{\delta}_{peak}$. The continuous propagation of the rupture front is achieved due to the spatially larger $\Delta\tau_d$ compared to the self-arresting rupture, which increases the ratio $\Delta\tau_d/\Delta\tau_b$. This yields a sustained crack-like rupture, implying that after the passage of the rupture front, all points behind the rupture tip continue slipping with $\dot{\delta}$ that converges to a constant residual value (see Supplementary Material).

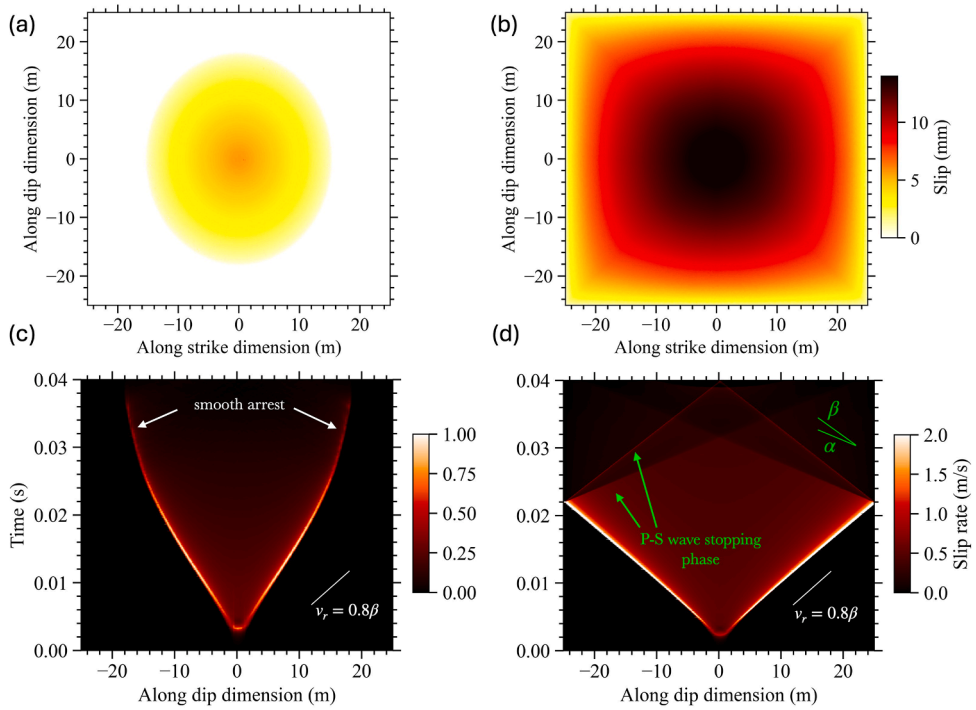


Fig. 2. Evolution of dynamic rupture models. a-b) Final slip distribution for the self-arresting and run-away ruptures, respectively. c-d) Evolution in time of points along fault dip for self-arresting and run-away rupture models. The colormap indicates the fault point's slip velocity.

In this scenario, the rupture does not arrest spontaneously but instead stops only when the crack tip encounters the geometrical barriers imposed by the finite fault size. This interaction generates multiple stopping phases, which have not yet been observed in natural earthquakes, but is consistent with previous studies (Madariaga, 1976; Kaneko and Shearer, 2014). The reflections from the fault boundaries, propagating backward at α and β wave speeds, as illustrated in Fig. 2d, introduce additional interference in the radiated waveforms, further increasing the complexity of the released moment at each fault point, beyond what is solely associated with the passage of the primary rupture front.

2.4. Synthetic waveforms

To what extent can self-arresting and runaway ruptures be distinguished by their radiated seismic wavefields? We address this by computing synthetic waveforms from dynamic rupture simulations and applying spectral inversion.

Our analysis focuses on receivers at depth, mimicking underground laboratory settings (take-off angle ~ 90°) as at the BULGG site (Mesimeri et al., 2025). The 60°-dipping normal fault and this geometry produce a P- and S-wave energy distribution at the seismograph that differs from that of natural earthquakes recorded at the surface. Fig. 3 shows ground velocity components versus azimuth for self-arresting (top) and runaway (bottom) ruptures. The orthogonal (a, d) and parallel (b, e) components contain both P- and S-wave phases, while the vertical components (c, f) capture only S-wave energy. Unlike natural settings, the near-straight ray paths, due to short distance (850 m), the receivers depth and the constant waves-velocity model at BULGG (Obermann et al., 2024) produce this distinct polarization pattern. Therefore, we rotate the seismograms into L (P-wave), Q (SV-wave), and T (SH-wave) directions to isolate P- and S-wave energy for spectral inversion, and then perform a single-component inversion for the P-wave and a two-component inversion for the S-waves.

Runaway ruptures generate sharper, more peaked P- and S-wave phases and richer high-frequency content with post-peak oscillations, while self-arresting ruptures exhibit smoother decay, suggesting lower high-frequency radiation than expected.

3. Spectral inversion method

In order to infer the source behavior from the waveforms, we use the probabilistic approach proposed by Supino et al. (2019) to estimate the earthquake source parameters: seismic moment M_0, corner frequency f_c and high-frequency decay factor γ. We invert the displacement amplitude spectra of synthetic seismograms generated by dynamic rupture models (Fig. 4), assuming the far-field approximation (Madariaga and Schubert, 2015). Both P and S waves are inverted separately, modeling the displacement spectrum as the convolution of the source time function and the Green's propagator, $u = S \cdot G$, where S represents the source time function and G the Green's operator. We assume a generalized (Brune, 1970) source spectrum model:

$$S(M_0, f_c, \gamma; f) = \frac{M_0}{1 + (\frac{f}{f_c})^{\gamma}} \tag{1}$$

where f is the frequency. We adopt the Green's functions from Aki and Richards (2002), considering geometric spreading and neglecting anelastic attenuation and site amplification due to the elastic assumption of the dynamic models with receivers at depth. The seismic moment M_0 is retrieved from the low-frequency limit ($f \to 0$) of the displacement spectrum (Ω_0):

$$M_0 = \frac{4\pi\rho r c_r^{1/2} c_s^{5/2} \Omega_0}{R_c F_s} \tag{2}$$

where r is the source-receiver distance, c_r and c_s are the wave velocities at the station and source, R_c the radiation pattern coefficient, F_s the free surface coefficient.

For each station, we estimate the joint a-posteriori probability density function (PDF) of source parameters $\sigma(\mathbf{m} = \log M_0, f_c, \gamma)$, combining the a-priori PDFs over model and data spaces and the likelihood of the

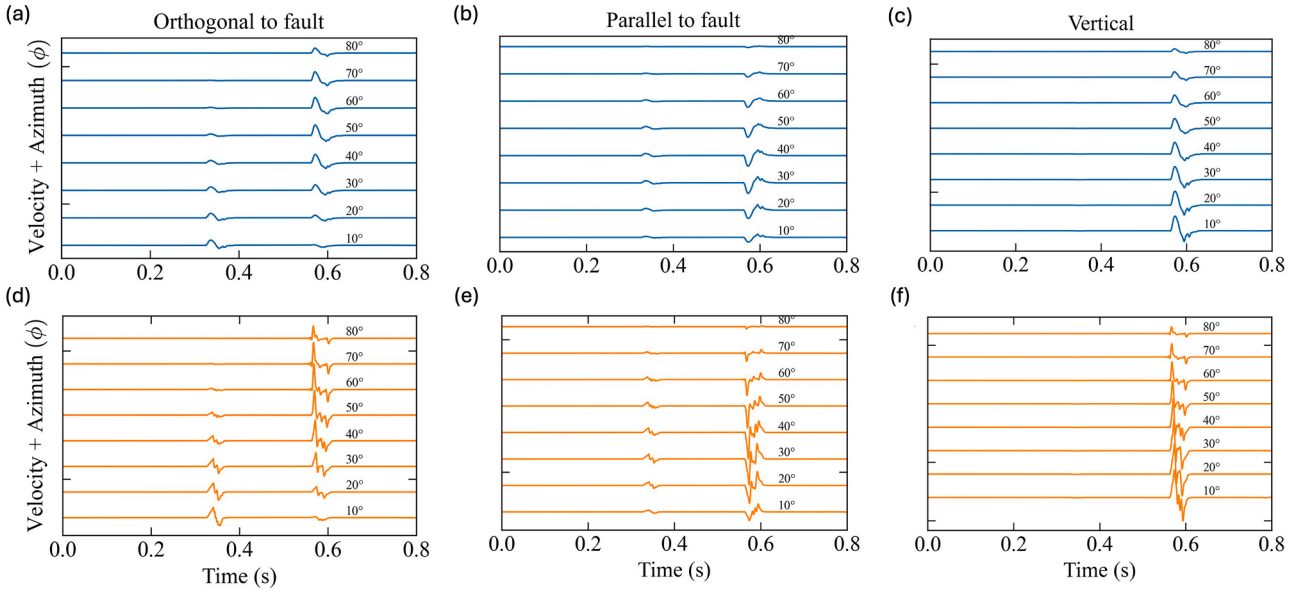


Fig. 3. Waveforms for the self-arresting and run-away rupture scenario for receivers with 90° take of angle. Waveforms represent particle velocity recorded at each seismograph for three components: (a-d) fault-perpendicular, (b-e) fault-parallel, and (c-f) vertical. In all panels, the waveforms are vertically shifted according to the azimuthal angle ϕ, which increases from bottom to top.

data given the assumed model. For each parameter, the solution and related uncertainty are the mean and standard deviation of the corresponding marginal PDF, respectively. This uncertainty accounts for the correlation existing between the source parameters as it comes from the integration of the joint PDF. Overall, a solution quality is evaluated by comparing the marginal PDF with a Gaussian distribution, rejecting solutions that do not pass the predefined quality threshold (Supino et al., 2019).

4. Source parameters from spectral inversion

We perform a probabilistic inversion of the Fourier displacement amplitude spectra from synthetic P- and S-waves generated by the self-arresting and run-away rupture models described above. The inversion allows us to retrieve key source parameters, starting with the seismic moment (M_0), and to explore differences in high-frequency spectral behavior between the two rupture types. Fig. 4 illustrates an example of spectral fits for both rupture scenarios at a representative receiver ($\phi = 40°$, take-off angle 90°), separately for P- and S-waves.

4.1. Seismic moment estimation

Estimating the seismic moment M_0 is essential in induced seismicity studies, as it allows computing the moment magnitude M_w and constraining earthquake size, particularly for small events where M_L scaling is less reliable (Deichmann, 2017). We compute M_0 for both decelerating and accelerating ruptures using Eq. (2), and derive M_w using the relation proposed by Kanamori (1977), with M_0 expressed in [Nm].

$$M_w = \frac{2}{3}\log_{10} M_0 - 6.07 \tag{3}$$

Fig. 5a-b shows M_w values obtained by fitting the displacement spectra (see Fig. 4a-b) of the self-arresting rupture model for each station at different ϕ. Gray triangles in panels (a) and (b) display the M_w values obtained by inverting P- and S-waves, respectively, adopting a constant radiation pattern coefficient (grey horizontal line in panels c and d), as widely used in literature (Boore and Boatwright, 1984). Black solid lines in Fig. 5a-b represent the real moment magnitude value ($M_w = 0.76$). The figure shows a large variability of M_w with the receiver's azimuth, ranging from $-0.9 \leq M_w \leq 0.85$. The largest deviations of estimated M_w are at high azimuthal angles ($\phi \geq 45°$) for both P- and S-waves, with S-wave magnitudes always underestimating the real value. Similar results

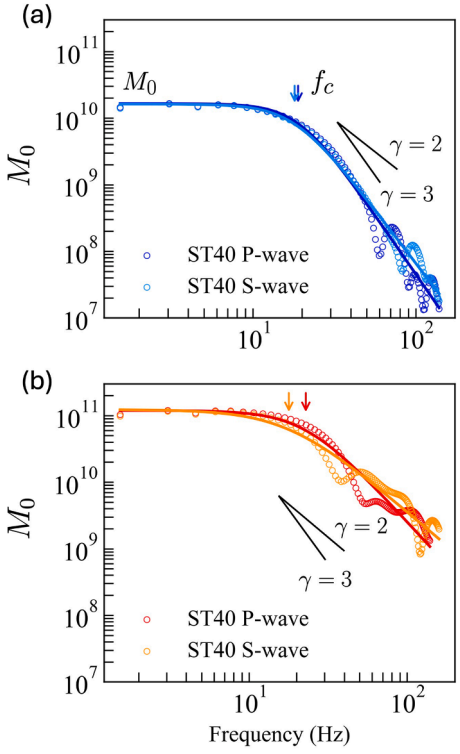


Fig. 4. Spectrum of self-arresting and run-away rupture models at a station located at $\phi = 40°$ and 90° take-off angle. Panel (a) shows the observed spectra for the self-arresting rupture (dark blue circles for the P-wave, light blue circles for the S-wave), the corresponding best fit spectra (solid lines) and estimated corner frequencies (vertical arrows). Panel (b) as for panel (a) for the run-away rupture, with P- and S-waves in red and orange, respectively. Reference slopes for $\gamma = 2, 3$ are shown in both panels.

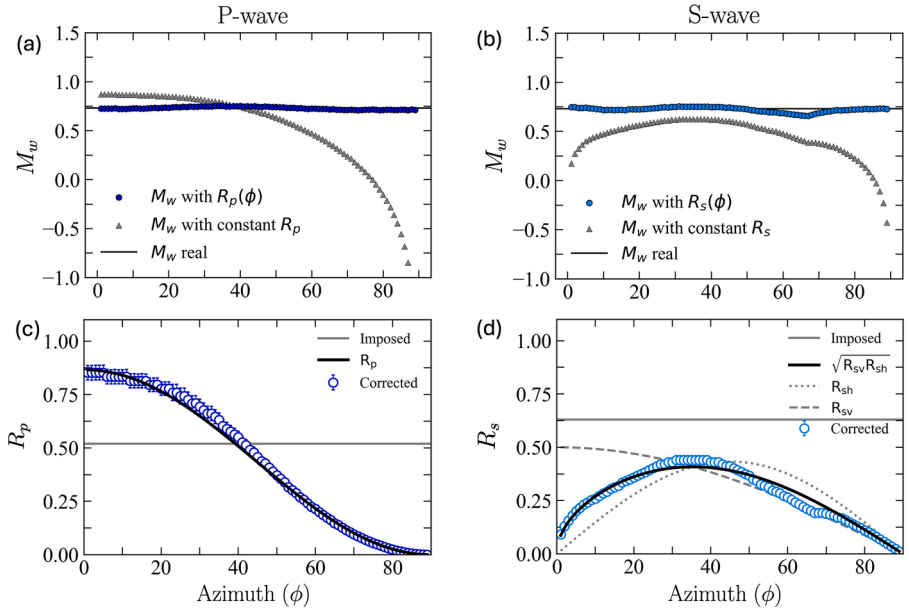


Fig. 5. Moment magnitude (M_w) estimation for a self-arresting rupture as a function of ϕ with a fixed take of angle of 90°. (a) Estimated M_w from P-waves with constant radiation pattern, gray triangles. Blue dots shows M_w with theoretical radiation pattern correction ($R_p(\phi)$). (b) Estimated M_w from S-waves with constant radiation pattern, gray triangles. Cyan dots shows M_w with theoretical radiation pattern correction ($R_s(\phi)$). The black solid line represent the "real" M_w of the forward model. (c-d) Black solid lines indicate the theoretically derived R_p for P-waves and the geometrical average (between R_{SV} and R_{SH}) R_s, for the S-wave. Solid gray lines represent the assumed constant R_c. The empirically corrected R_c, required to match real M_w, are shown with blue and cyan dots, corresponding to P- and S-waves, respectively. In panel (d), the gray dotted line corresponds to R_{SH} and the gray dashed line to R_{SV}.

have been obtained for the run-away rupture model as shown in Figure S4 (Supplementary Materials).

The azimuthal variation of estimated seismic moment arises from the radiation pattern R_c in (2). Assuming a constant radiation pattern at this scale could introduce errors up to a factor 2 in the case of P-waves (Fig. 5a). This implies that a correction is necessary to estimate the moment magnitude. Following Udías et al. (2014), the radiation pattern coefficient R_c can be expressed in terms of the azimuthal angle ϕ, the take-off angle i, the fault dip and the rake angle λ. Under the assumption of a point source, this enables the derivation of the theoretical radiation pattern for the adopted source-to-receiver geometry for the P-wave (R_P), SH- (R_{SH}) and SV-waves (R_{SV}) as well as the average value for the whole S-wave field, R_s, (see panels Fig. 5c and d). Since the true M_w is known, we can compute the empirical R_c factor required to match the estimated M_w values to the actual one. It can be observed that the theoretical R_c (solid black lines) closely matches the empirically corrected R_c (open circles in panels Fig. 5c and d) for both P- and for S-waves. It is worth noting that, in the case of the S-wave inversion, the black solid line represents the geometric mean of the SH and SV wavefields (as done during the inversion procedure). The inferred M_w values computed by correcting for theoretical R_c (blue dots in Fig. 5a and cyan dots in Fig. 5b,) shows excellent agreement with the constant ground truth magnitude. We remark that, given this source-receiver configuration, the azimuthal correction at low ϕ angles is primarily limited by low R_{SH} values, despite the SV radiation being dominant in the wavefield. Our results highlight the dispersion in M_w estimates that arises when using a constant R_c, and the importance of applying the appropriate theoretical radiation pattern correction.

4.2. Corner frequency and high-frequency decay factor

The corner frequency (f_c) is one of the most important observables in seismology (Brune, 1970; Allmann and Shearer, 2009; Schliwa and Gabriel, 2023). Its main feature is that f_c is related to the equivalent radius of the earthquake source (r), which is not directly measurable from recorded data. Therefore, f_c is commonly used to estimate r for different source models (Hanks and Wyss, 1972). Brune's model (Brune, 1970) kinematically describes a circular crack with instantaneous stress drop, while alternative kinematic or dynamic rupture models incorporate finite rupture velocity and stress evolution (Sato and Hirasawa, 1973; Madariaga, 1976; Kaneko and Shearer, 2014; Wang and Day, 2017). Despite shared idealized assumptions, their differing physics yield variations in $f_c - r$ scaling and high-frequency spectral decay.

Fig. 4 shows displacement spectral fits for P- and S-waves in the self-arresting (a) and runaway (b) rupture models at $\phi = 40°$ and a take-off angle of 90°. The fits are of high quality and highlight key features. In the self-arresting case (Fig. 4a), P- and S-wave spectra and their corner frequencies (f_c^P, f_c^S) are nearly identical, indicating no corner frequency shift (Dahlen, 1974). In contrast, the runaway rupture (Fig. 4b) shows the expected difference with $f_c^P > f_c^S$. The high-frequency decay is steeper for the self-arresting rupture, indicating reduced high-frequency radiation.

To further investigate these features, Fig. 6 summarizes the inferred γ (top panels) and f_c (bottom panels) for all receivers at 90° take-off angle. The results are tightly clustered around their mean values. A clear difference emerges, self-arresting ruptures exhibit systematically larger high frequency decay with average values of $\overline{\gamma}_P = 3.73 \pm 0.02$ and $\overline{\gamma}_S = 3.44 \pm 0.03$ (Fig. 6c) and a near-unity ratio $\overline{\gamma}_P/\overline{\gamma}_S = 1.08$. In contrast, runaway ruptures produce spectra with smaller high-frequency decay, as shown by their average decay values: $\overline{\gamma}_P = 2.39 \pm 0.04$ and $\overline{\gamma}_S = 1.97 \pm 0.02$ (Fig. 6a-c).

The distributions of inverted f_c values for both P- and S-waves are shown in Fig. 6d-e. For self-arresting rupture the mean values are $\overline{f_c}^P = 20.39 \pm 0.11$ Hz and $\overline{f_c}^S = 19.94 \pm 0.21$ Hz, and their ratio is $\overline{f_c}^P/\overline{f_c}^S \approx 1$, confirming the lack of corner frequency shift. For runaway rupture the mean values are $\overline{f_c}^P = 19.31 \pm 0.12$ Hz and $\overline{f_c}^S = 14.47 \pm 0.13$ Hz, and their ratio is $\overline{f_c}^P/\overline{f_c}^S \approx 1.3$. Fig. 6f illustrates the spread of values around the mean, which arises from both the azimuthal dependence of the radiated wavefield and the inherent uncertainties of the inversion process. A slightly greater dispersion is observed for the S-wave f_c compared to the estimates from P-waves. As expected for a canonical accelerating crack-

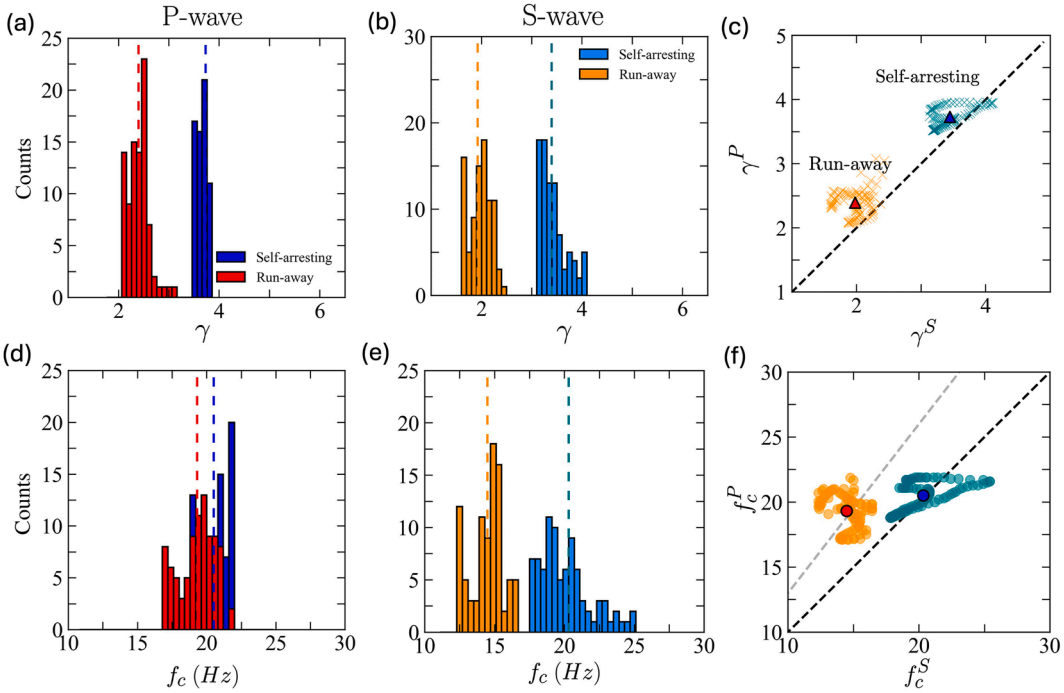


Fig. 6. Corner-frequency and high-frequency fall-off histograms for stations with variable ϕ and fixed take of angle at 90°. (a-d) Estimates of γ and f_c from P-wave inversion. Blue histograms correspond to the self-arresting model, while red histograms represent the run-away model. (c) Comparison of γ_p and γ_s for the self-arresting model (dark cyan) and the run-away model (orange). The dashed black line denotes the 1:1 relationship. (b-e) γ and f_c from S-wave inversion, with self-arresting results shown in cyan and run-away in orange. (f) Comparison between f_c^P and f_c^S for the self-arresting model (dark cyan) and the run-away model (orange). The dashed black line denotes the 1:1 relationship, while the gray dashed the 1:3.

like model, the P-wave corner frequency $\overline{f_c}^P$ retrieved for the runaway model is greater that the S-wave $\overline{f_c}^S$.

A more detailed analysis of the spectral inversion results is presented in Fig. 7, where the f_c and the γ are plotted as a function of the azimuthal angle (ϕ). Since our dynamic rupture models provide a known source radius r, we can directly compute the dimensionless parameter $k = f_c r/\beta$ originally defined by Brune (1970), where β denotes the shear-wave velocity. This parameter is used in the literature (Kaneko and Shearer, 2014, and references therein) to infer the source radius from f_c estimates and the commonly adopted values are listed in Table S1 (Supplementary Materials). Panels (a, b) refers to the self-arresting model, and(c, d) to the runaway case. The largest azimuthal variations of γ occur in S-waves with standard deviations of 0.26 (self-arresting) and 0.21 (runaway), compared to 0.15 and 0.12 for P-waves. The self-arresting model shows a more coherent azimuthal pattern and stronger high-frequency decay than the runaway case, which follows the classic ω^2 behavior. Overall, the azimuthal variability of γ is lower than in models with abrupt rupture arrest at barriers (e.g., Kaneko and Shearer, 2014; Lin et al., 2025). Panels (b, d) or Fig. 7 show that k values are generally lower for self-arresting ruptures ($\overline{k}_P = \overline{k}_S = 0.24$) than for runaway ruptures ($\overline{k}_P = 0.36$, $\overline{k}_S = 0.27$), reflecting their lower rupture velocities. Including all receivers (considering all the take-off angles) yields lower average spherical k values, consistent with Kaneko and Shearer (2015), enabling robust comparison. For runaway ruptures we find $k_p^{\text{sphere}} = 0.32$ and $k_s^{\text{sphere}} = 0.24$, while self-arresting ruptures converge to $k_p^{\text{sphere}} = k_s^{\text{sphere}} = 0.23$.

Importantly, this variation in k reflects rupture velocity, not moment magnitude, since the actual source radius is used in the calculation. Moreover, this result confirms the lack of a corner frequency shift between P- and S-waves for self-arresting ruptures also at different azimuth. While k_P aligns well with trends inferred in the literature (decreasing with v_r/β), k_S appears to be similar for run-aways and self-arresting ruptures (Table S1). In contrast, the values inferred for runaway ruptures closely match those of Kaneko and Shearer (2014), whose cohesive-zone models also feature rupture arrest at barriers. It is important to recall that self-arresting model represents a decelerating rupture with variable speed (v_r), gradually arresting at $\overline{v}_r = 0.2\beta$. In contrast, the runaway rupture features continuous tip acceleration, reaching $\overline{v}_r = 0.76\beta$ before abrupt arrest from geometrical constraints, generating multiple stopping phases.

Fig. 7 shows consistent trends of f_c and γ versus ϕ, confirming their well-known trade-off in spectral inversion. Our probabilistic approach (Supino et al., 2019) addresses this limitation by constructing marginal PDFs and automatically verifying that the solutions are well constrained (Figure S3). These robust results support our conclusion that self-arresting ruptures exhibit stronger high-frequency decay and no P-S f_c shift.

5. Discussion

5.1. High-frequency radiation

A key difference between self-arresting and run-away rupture models lies in the high-frequency content of the radiated wavefield. Self-arresting ruptures show a marked depletion of high frequencies ($\gamma \geq 3$) compared to the classic omega-squared model ($\gamma = 2$) typically observed for run-away ruptures. While high-frequency content can originate from both source complexity and crustal heterogeneity, we isolate the source contribution by modeling a homogeneous elastic medium without anelastic attenuation, accounting only for geometrical spreading. This allows a direct interpretation of high-frequency differences in terms of rupture dynamics, being aware that the potential application to real earthquake data will require accounting for the strong correlation between the source and anelastic attenuation characterizing observed high-frequency decays (e.g., Shearer et al., 2024).

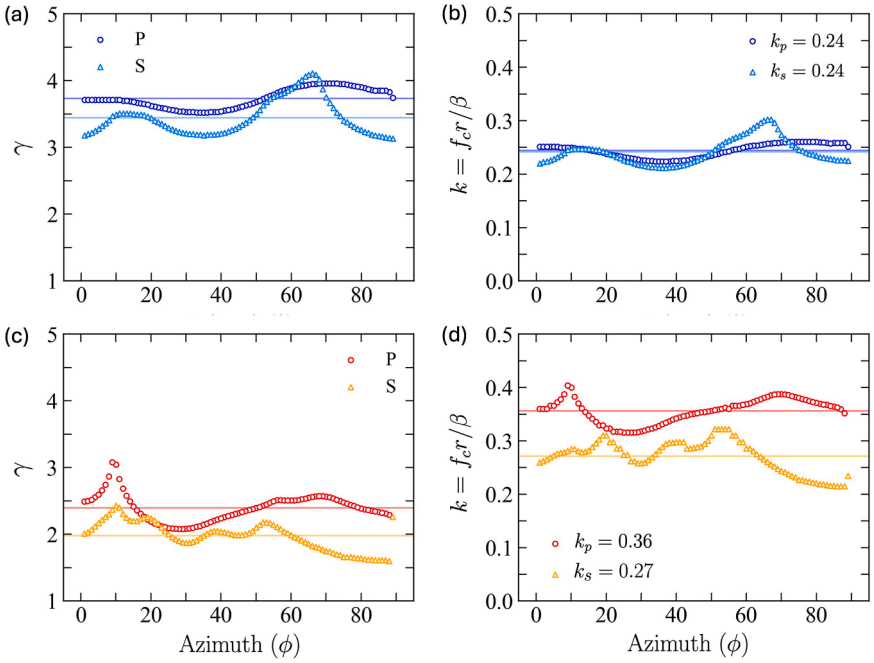


Fig. 7. Results of γ and k as a function of azimuthal angle ϕ with a fixed 90° take of angle. (a-b) Self-arresting model: blue dots correspond to P-wave inversion, and cyan dots to S-wave inversion. (c-d) γ and k for the run-away rupture model: red dots represent P-wave inversion, while orange dots indicate S-wave inversion. The legend in panels (b) and (d) shows the mean k values obtained from P- and S-wave inversions for both the self-arresting and run-away rupture models. The horizontal lines in all the panels represent the mean values.

To explore this further, we analyze the moment rate function, $\dot{M}_0(t)$, for decelerating and accelerating ruptures, meaningful as we intentionally operate under far-field conditions.

The self-arresting (decelerating) rupture exhibits a bell-shaped moment rate, $\dot{M}_0(t)$, as shown in Fig. 8a, with a peak nearly symmetric with respect to the total rupture duration (Fig. 8c). The initial growth of $\dot{M}_0(t)$ corresponds to the early rupture phase, during which the expanding crack accelerates and the moment rate increases approximately quadratically ($\propto t^2$). This is followed by a phase of nearly constant v_r (Mosconi et al., 2025). Beyond the peak, the crack decelerates, leading to a gradual decrease in $\dot{M}_0(t)$. These smooth transitions shape the characteristic $\dot{M}_0(t)$ curve and control its spectral content, as seen in the spectrograms of moment acceleration, $\ddot{M}_0(t)$, in Fig. 8e. This behavior reflects the dynamic rupture model properties (continuously increasing normal stress and constant prestress) adopted to simulate a decelerating rupture front.

In contrast, the $\dot{M}_0(t)$ of the accelerating rupture (Fig. 8b) shows quadratic growth up to the peak (Ke et al., 2022) for $t < 0.02$ s, as indicated by the black dotted line. This reflects an increasing $\dot{\delta}$ as the crack expands. The finite fault geometry ultimately limits rupture growth. When the rupture front reaches the fault boundary, it stops abruptly, causing a sudden drop in $\dot{M}_0(t)$ at its peak, as illustrated by $\ddot{M}_0(t)$ in Fig. 8d. This abrupt arrest radiates strong high-frequency energy, clearly visible in the spectrograms of $\ddot{M}_0(t)$ (Fig. 8f). The high-frequency burst coincides with the collision of the rupture front with the fault boundary, followed by a second burst at $t \approx 0.03$ s due to reflected phases (Fig. 2e). Such high-frequency radiation often localizes in areas of sharp fault-zone property contrasts, as observed during the 2000 Tottori earthquake, where most energy came from a small part of the fault with high stress drop and strength excess (Pulido and Dalguer, 2009).

What is observed from the on-fault $\dot{M}_0(t)$ is in good agreement with the results of the spectral inversion. The run-away rupture exhibits a larger high-frequency content, with lower γ values compared to the decelerating rupture (Fig. 6). A key insight from the on-fault $\dot{M}_0$ is that high-frequency radiation occurs only during specific rupture phases, predominantly governed by rupture arrest (Madariaga, 1976; Ripperger et al., 2007), which fundamentally shapes the coherence of the widely observed ω^2 model. In contrast, self-arresting ruptures, characterized by a gradual deceleration and no abrupt stopping phase, suppress the excitation of high-frequency waves, leading to significantly larger γ values.

5.2. On the P- to S-wave corner frequency shift

The differing durations of the P- and S-wave pulses (and consequently their different f_c values) are well-known features, observed in both natural earthquakes and theoretical source models (Molnar et al., 1973; Kaneko and Shearer, 2014). In our simulations, however, the decelerating rupture exhibit similar corner frequencies ($f_c^P \approx f_c^S$), whereas the run-away (or accelerating) rupture shows a noticeable shift with $f_c^P / f_c^S \approx 1.3$ (see Fig. 6). This opens the question of how frequently self-arresting ruptures, if they exist, occur in nature, whether they represent unconventional rupture scenarios, or if the lack of observed stopping phases makes them more common than typically assumed. Nevertheless, while most natural earthquake estimates show $f_c^P > f_c^S$, some degree of variability exists, including cases where $f_c^P \approx f_c^S$ (Abercrombie, 1995; Godano et al., 2015).

To further validate our f_c estimates, Fig. 9 shows the P- and S-wave pulses for the self-arresting and run-away ruptures (panels a and b, respectively). Waveforms are aligned to their arrival times and normalized by their peak amplitudes to enable direct comparison of pulse durations. As expected, the self-arresting rupture exhibits similar P- and S-wave durations (Fig. 9a), supporting the spectral inversion results and confirming the link between source duration and f_c. In contrast, the S-wave pulse of the run-away rupture is clearly longer (Fig. 9b), consistent with $f_c^S < f_c^P$.

To understand the mechanism behind the different duration of the radiated P- and S-waves of decelerating and accelerating ruptures, we apply a simplified description based on the following relation to estimate the arrival times of two stopping phases characterizing the arrest stage of an accelerating rupture that encounters a fault geometrical barrier (modified from Madariaga, 1976; Wang and Day, 2017):

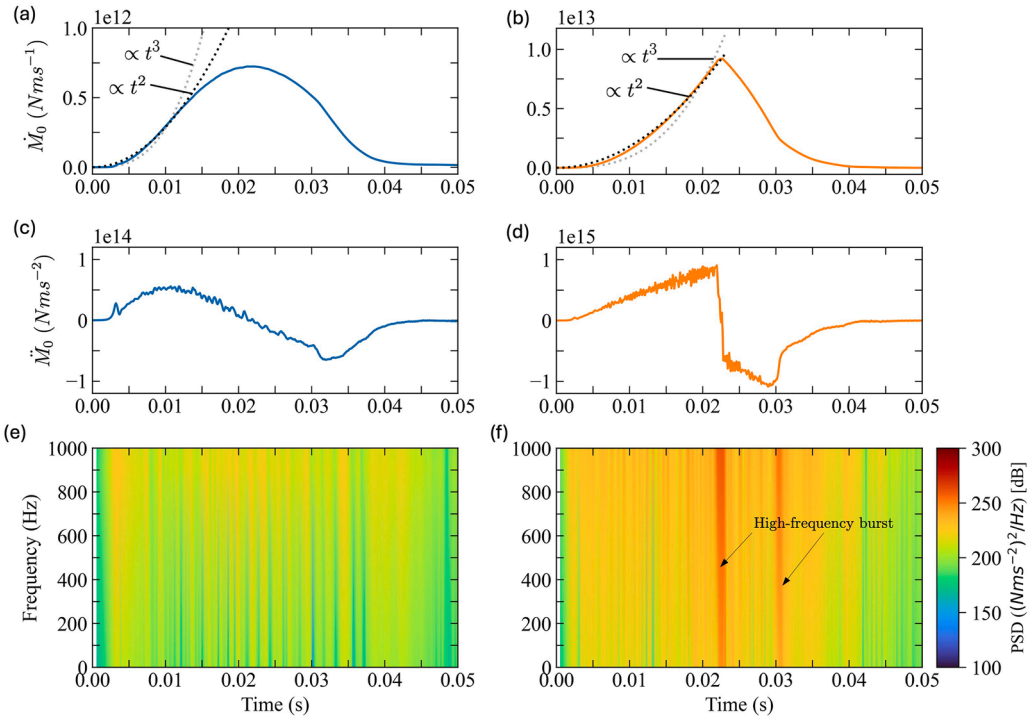


Fig. 8. Moment rate function ($\dot{M}_0(t)$) of dynamic source models. $\dot{M}_0$ for the self-arresting (a) and accelerating rupture models (b); the black dashed lines represent the quadratic fitting function. Second time derivative of the moment, $\ddot{M}_0(t)$, for the self-arresting (c) and run-away models (d). Spectrograms of $\ddot{M}_0(t)$ for the self-arresting (e) and run-away models (f).

$$\begin{cases} t_i = r\left(\frac{1}{v_r} - \frac{\sin\phi}{c}\right) + \frac{x}{c} \\ t_f = r\left(\frac{1}{v_r} + \frac{1}{c}\right) + \frac{x}{c} \end{cases} \tag{4}$$

where c is the wave speed, x the source to receiver distance from the hypocenter (i.e., the distance under the point source approximation), ϕ is the azimuth in our geometry, and r the source radius. The times t_i correspond to the arrival of the stopping phase radiated from the nearest fault edge, while t_f marks the arrival of waves emitted by the last points to stop slipping, which coincide with the hypocenter. In other words, t_f corresponds to the arrival of the back-propagating stopping phase at the center of the fault. When the rupture front encounters a "barrier," which may result from a strong stress gradient, a variation in fault strength, or, as in this case, a geometrical discontinuity such as the fault boundaries, it arrests abruptly. This abrupt arrest produces a sudden drop in the moment rate ($\dot{M}_0$) and its derivative ($\ddot{M}_0$), as shown in Fig. 8b. This generates the first stopping phase, originating from the closest points on the fault edge and arriving at time t_i, indicated in the figure by gray and black lines for the P- and S-waves, respectively.

The actual termination of slip occurs at the center of the fault, which is the last point to stop radiating. Consequently, the real duration of the radiated pulse, as seen at the receiver, is controlled by the waves traveling from the center of the fault to the receiver. After the front reaches the barrier, reflected fronts propagate backward from the edge of the fault at α and β speeds (Madariaga, 1976), as illustrated in Fig. 2e. The difference in arrival times of these secondary stopping phases for P and S waves (marked by gray and black bars in Fig. 9b) explains the longer duration of the S-wave pulse compared to the P-wave.

In contrast, self-arresting ruptures do not generate stopping phases. The rupture front slows down progressively due to the decreasing ratio $\Delta\tau_d/\Delta\tau_b$ and the spatially increasing fracture energy G_c, ultimately leading to spontaneous arrest. This deceleration is accompanied by a reduction in $\dot{\delta}_{peak}$ and, importantly, a decrease in the residual $\dot{\delta}$ behind the rupture front, which further limits high-frequency radiation. Unlike in the runaway case, in this model slip ceases earlier at the center and later at the edges of the rupture front, further emphasizing the absence of a distinct stopping phase and the progressive nature of rupture arrest.

5.3. Stress drop estimation

In our fluid-induced rupture simulations, the on-fault static stress drop, here denoted as $\Delta\tau_s$, varies spatially due to non-uniform effective normal stress conditions (σ_n'). This spatial variability complicates the definition of a single, representative static stress drop value.

Fig. 10 compares the spectral stress drop estimated from synthetic waveform inversions, $\Delta\tau_{\text{inversion}}$, with several definitions of the "true" static stress drop, derived directly from the rupture simulations. To estimate stress drop from spectral inversion, we apply the classical (Eshelby, 1957) formula:

$$\Delta\tau = \frac{7}{16}\frac{M_0}{r^3} \tag{5}$$

For $\Delta\tau_{\text{inversion}}$, both M_0 and r are obtained from the spectral inversion of synthetic seismograms, representing what an observer would infer from seismic data. Notably, M_0 is initially computed assuming a constant radiation pattern, introducing a systematic error in the estimates (Section 4.1).

For comparison, we also compute a "ground truth" stress drop, $\Delta\tau_M$, by applying Eq. (5) using the exact M_0 and r extracted directly from the dynamic rupture simulations. Given the non-uniformity of $\Delta\tau_s$ along the fault, these scalar estimates should be interpreted as slip-weighted averages (Madariaga, 1979).

For completeness, we consider two additional definitions of static stress drop based on its spatial distribution on the fault. The first is the area-averaged stress drop, $\Delta\tau_A$, defined as:

$$\Delta\tau_A = \frac{\int_\Sigma \Delta\tau\, dS}{A} \tag{6}$$

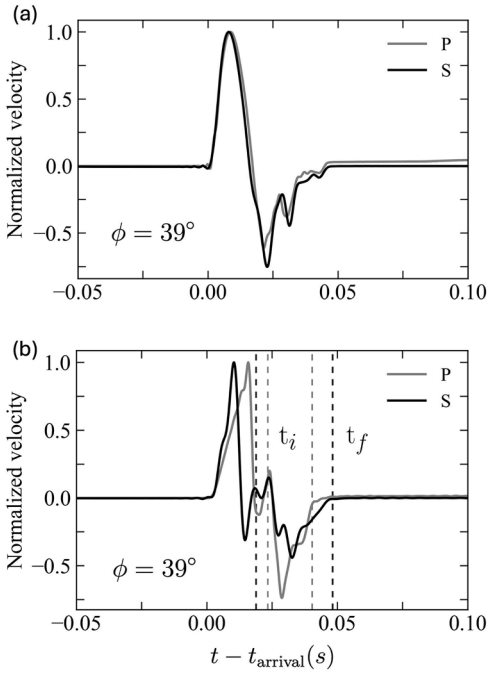


Fig. 9. Normalized particle velocity for seismographs located at 850 m distance, $\phi = 39°$ and 90° take of angle. (a) Self-arresting models, (b) Run-away rupture, where the diamond indicates the arrival time of the first stopping phase, and the triangle indicates the arrival time of the second stopping phase. The blue color refers to the P-wave, while the red corresponds to the S-wave. In both panels, gray lines represent the P-wave pulses, while the black line indicates the S-wave.

and the second is the slip-weighted (or energy-based) stress drop, $\Delta\tau_E$, given by:

$$\Delta\tau_E = \frac{\int_\Sigma \Delta\tau \cdot \Delta\delta \, dS}{\int_\Sigma \Delta\delta \, dS} \tag{7}$$

where $\Delta\delta$ is the final slip at each point on the fault (Noda et al., 2013). This variety of definitions underscores the inherent difficulty in associating the spectral stress drop with a unique physical quantity on the fault, and highlights the need for caution when interpreting spectral inversion results. Despite their conceptual differences, all three definitions yield comparable values. Averaging across them, we obtain $\Delta\tau_s = 1.33$ MPa for the self-arresting rupture and $\Delta\tau_s = 2.48$ MPa for the runaway rupture (Fig. 10a, d). This suggests that the overall rupture shape has limited impact on the estimation of $\Delta\tau$, relative to the influence of local slip heterogeneities (Noda et al., 2013).

Due to the use of a linear slip-weakening friction law, and specifically because of the absence of stopping phases in the self-arresting rupture, static and dynamic stress drops coincide locally. In contrast, runaway ruptures are characterized by stress overshoot (Figure S2, supplementary material), resulting from back-propagating stopping phases. In these cases, static and dynamic stress drops differ significantly. Specifically, the average apparent stress drop $\Delta\tau_A$ reaches 2.5 MPa when accounting for the dynamic overshoot, while the expected value imposed by the constitutive law (i.e., neglecting the overshoot) is 2.0 MPa. This discrepancy indicates that, for runaway ruptures that arrest at a barrier (geometrical in this case), the estimated $\Delta\tau_{\text{inversion}}$ implicitly includes an additional stress drop component. This component is linked to the overshoot that occurs over the same rupture area due to rupture front reflections (stopping phases). As a consequence, only in the case of decelerating ruptures that do not exhibit stress overshoot, the stress drop retrieved via spectral inversion corresponds to the actual $\Delta\tau_s$ associated with the propagating rupture front.

Another critical factor in reconciling spectral and static stress drop estimates is the azimuthal variability of the radiation pattern coefficient R_c. If R_c is not accounted for, stress drop estimates exhibit substantial scatter (gray histogram bars in Fig. 10) due to the directional dependence of the radiation pattern and its impact on M_0 measurements. Incorporating azimuth-dependent R_c (Fig. 5) significantly reduces this scatter and aligns spectral estimates more closely with the static $\Delta\tau_A$ (colored histogram bars in Fig. 10), especially for S-waves (Fig. 10c, f).

Neglecting this correction results in systematic underestimation of $\Delta\tau_{\text{inversion}}$–by up to 50% for S-waves in self-arresting ruptures and 40% in runaway events. For P-waves, the bias is less pronounced only when azimuthal coverage is dense, enabling an implicit compensation. However, such coverage is rarely guaranteed in real-world deployments. These findings highlight the need for incorporating radiation pattern corrections even at small magnitudes ($M_w \sim 1$), to obtain physically consistent estimates of source parameters.

5.4. Potential applications to real earthquake data

Our numerical study establishes a framework that could potentially enable the identification of distinct rupture behaviors, self-arresting versus runaway, from real earthquake waveforms.

Observational evidence of $f_c^P \approx f_c^S$ has been reported in various contexts (Abercrombie, 1995; Godano et al., 2015). High-frequency fall-off rates (γ) can span a broad range of values, with $\gamma > 1.5$ being the physical bound to have convergence in the radiated energy, and $\gamma \geq 3$ systematically observed in specific tectonic settings (e.g., for slow earthquakes in Japan by Supino et al., 2020). Our results provide a new perspective for interpreting these observations, suggesting that some of them may reflect the spectral signature of self-arresting ruptures.

In this study, we could easily isolate source-related characteristics in the displacement spectra by modeling a homogeneous elastic propagation medium (and an embedded planar fault). However, the analysis of real earthquakes will require a spectral inversion framework where both the anelastic attenuation factor (Q) and the high-frequency fall-off rate (γ) are considered as free parameters. We note that, to date, the source high-frequency decay is often a fixed parameter in spectral inversions and the omega-squared model (Brune, 1970) is assumed. For instance, in the recent community study by Abercrombie et al. (2025), only 3 among the 56 unique submissions from 20 different research groups used γ as a free parameter in the modeling.

The probabilistic framework adopted in this study is capable to obtain the joint PDF in a 4D model space that includes both Q and γ (Supino et al., 2019), as shown in a recent study from real earthquake displacement spectra (Supino et al., 2024). Alternatively, the well-known trade-offs among f_c, Q, and γ can be mitigated by adopting techniques such as spectral ratio analysis (Uchida et al., 2007), albeit some limitations exist also in this case (Abercrombie, 2015), in particular when investigating micro-earthquakes. Another promising approach for distinguishing rupture types lies in the analysis of earthquake source time functions (STFs) from near-field observations, provided that path effects are properly accounted - for instance, through empirical Green's functions (e.g., Oates et al., 2024).

The rapidly growing deployment of Near Fault Observatories (NFOs) for natural earthquakes –such as the NFO along the Alto Tiberina Fault System in Northern Apennines, Italy, (Chiaraluce et al., 2014; Poggiali et al., 2025)– and Geoenergy testbeds for induced seismicity –such as the BULGG underground laboratory in the Swiss Alps (Plenkers et al., 2023)– offers new opportunities to collect earthquake data recorded at close distances, over a broad frequency and magnitude range. Faults with creeping sections, due to the local variation of their stress field (Dreger et al., 2007), and fluid-induced seismicity represent suitable settings in which to search for self-arresting ruptures in future studies.

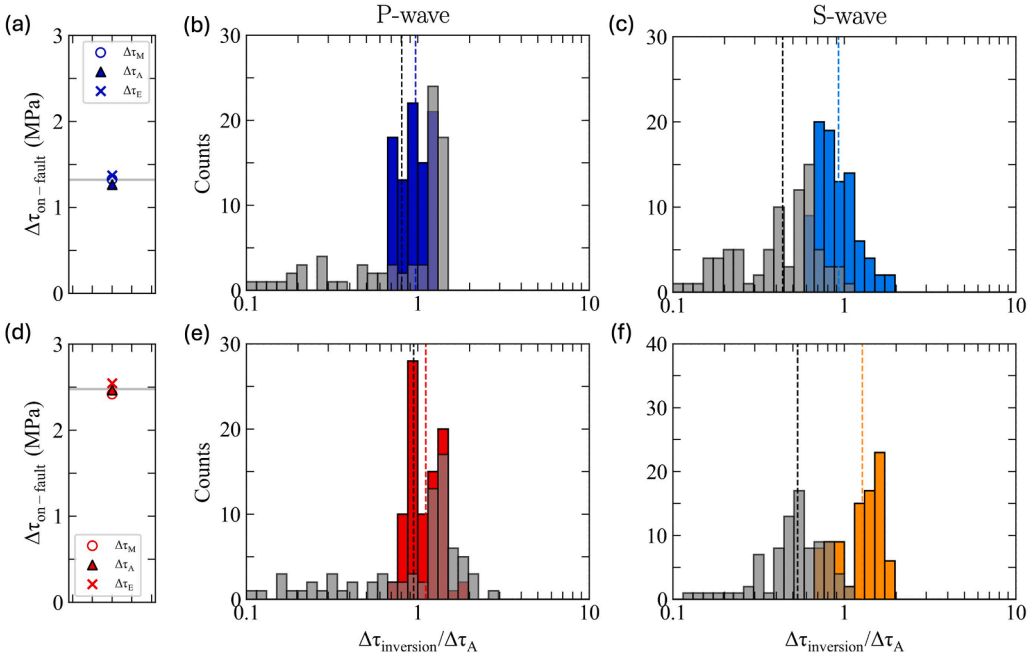


Fig. 10. Analysis of stress drop from spectral inversion and dynamic source models. Panels (a,d) show the average on-fault stress drop computed from dynamic rupture models using Eqs. (5)–(7), for both self-arresting (a) and runaway rupture scenarios (d). Panels (b,c) display histograms of the ratio between the estimated and the average on-fault stress drop ($\Delta\tau_{inversion}/\Delta\tau_A$) for P- and S-waves, respectively, in the self-arresting case. Panels (e,f) show the same ratio for the runaway rupture scenario. Gray histograms indicate stress drop estimates obtained without applying the correction for the azimuthally varying radiation pattern, colored histograms indicate estimates with applied azimuthal correction (see Section 4.1).

6. Conclusion

Understanding whether dynamic rupture spontaneously arrests or is limited by structural barriers remains a central question in earthquake mechanics. Here, we extend the concept of self-arresting ruptures (e.g., Xu et al., 2023) to fluid-induced micro-earthquakes that propagate beyond the nucleation zone, decelerate, and arrest spontaneously within fault boundaries, without invoking the traditional asperity-barrier model (Das and Aki, 1977).

In our 3D dynamic rupture simulations, runaway ruptures exhibit the classical ω^2 high-frequency decay and distinct f_c^P and f_c^S (Brune, 1970; Madariaga, 1976; Kaneko and Shearer, 2014). In contrast, self-arresting ruptures, characterized by a smooth moment rate function, suppress high-frequency radiation ($\gamma \geq 3$) and show no shift between f_c^P and f_c^S, reflecting the absence of abrupt stopping phases. These spectral features may serve as diagnostics for identifying self-arresting ruptures in real earthquakes, improving our understanding on how earthquakes stop.

The physical interpretation of the scaling parameter k, which relates the corner frequency to rupture dimensions, differs between self-arresting and runaway ruptures. For runaway events, k captures the combined effects of the forward-propagating rupture front and the backward-propagating stopping phases. This results in a longer source duration over the same rupture area and introduces a stress overshoot component associated with the stopping phase, which is implicitly included in the final stress drop estimate. In contrast, for self-arresting ruptures, k reflects only the propagation of the primary rupture front, as no stopping phase or stress overshoot occurs. Consequently, the inferred stress drop for self-arresting ruptures more accurately represents the dynamic stress drop associated to the forward-propagating rupture front.

Our findings demonstrate that k is sensitive to rupture dynamics, implying that corner frequency alone is an unreliable proxy for rupture size when rupture style is not accounted for. This study also highlights the importance of careful spectral inversion parameter selection and the usefulness of realistic rupture models to robustly identify diagnostic features.

CRediT authorship contribution statement

Francesco Mosconi: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization; **Elisa Tinti:** Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Conceptualization; **Mariano Supino:** Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation; **Alice-Agnes Gabriel:** Writing – review & editing, Validation, Supervision, Software, Resources, Methodology; **Emanuele Casarotti:** Writing – review & editing, Validation, Software, Methodology; **Men-Andrin Meier:** Writing – review & editing, Validation, Conceptualization; **Antonio Pio Rinaldi:** Writing – review & editing, Validation, Software, Resources, Methodology; **Domenico Giardini:** Validation, Project administration, Funding acquisition; **Massimo Cocco:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.epsl.2025.119816

Data availability

We use the SeisSol software package available on GitHub (https://github.com/SeisSol/SeisSol) to simulate all dynamic models. We use SeisSol, version 202,103_Sumatra-686-gf8e01a54 (master branch on commit dd018b3398258a23ec2a33c74bd7f31b503dcca6, v1.1.3-362-gdd018b33). The procedure to download and run the code is described in the SeisSol documentation (https://seissol.readthedocs.io/en/latest/). Downloading and compiling instructions are at https://

seissol.readthedocs.io/en/latest/build-seissol.html. Instructions for setting up and running simulations are at https://seissol.readthedocs.io/en/latest/configuration.html. Quickstart containerized installations and introductory materials are provided in the docker container and Jupyter Notebooks at https://github.com/SeisSol/Training. Example problems and model configuration files are provided at https://github.com/SeisSol/Examples, many of which reproduce the SCEC 3D Dynamic Rupture benchmark problems described at https://strike.scec.org/cvws/benchmark_descriptions.html.

All data required to reproduce the dynamic rupture scenarios are available at https://doi.org/10.5281/zenodo.14517701

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

I have nothing to declare If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank the Editor Dr. Jean-Philippe Avouac and the two anonymous reviewers for helpful comments and suggestions that improved the paper. We thank Ravil Dorozhinskii, Thomas Ulrich and the entire SeisSol team (www.seissol.org). We acknowledge the CINECA award under the ISCRA initiative, for the availability of high-performance computing resources and support.

This study has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme through the ERC-Synergy FEAR (Fault Activation and Earthquake Ruptures) project (Grant agreement 856559). FM, ET, MS, MAM, APR, DG and MC participated in this work within the activity plan of the ERC-Synergy project FEAR. MS acknowledges support from the Horizon Europe (HORIZON) project TRANSFORM2 (Grant agreement ID: 101188365) funded by the European Union. AAG acknowledges additional support from the National Science Foundation (NSF EAR-2225286, EAR-2121568, OAC-2139536, OAC-2311208), Horizon Europe (Geo-INQUIRE, grant number 101,058,518 and ChEESE-2P, grant number 101093038), and the National Aeronautics and Space Administration (NASA 80NSSC20K0495).

References

Abercrombie, R.E., 1995. Earthquake source scaling relationships from- 1 to 5 ML using seismograms recorded at 2.5-km depth. J. Geophys. Res. Solid Earth 100 (B12), 24015–24036.

Abercrombie, R.E., 2015. Investigating uncertainties in empirical green's function analysis of earthquake source parameters. J. Geophys. Res. Solid Earth 120 (6), 4263–4277.

Abercrombie, R.E., 2021. Resolution and uncertainties in estimates of earthquake stress drop and energy release. Philosophical Trans. R. Soc. A 379 (2196), 20200131.

Abercrombie, R.E., Baltay, A., Chu, S., Taira, T., Bindi, D., Boyd, O.S., Chen, X., Cochran, E.S., Devin, E., Dreger, D., Ellsworth, W., Fan, W., Harrington, R.M., Huang, Y., Kemna, K.B., Liu, M., Oth, A., Parker, G.A., Pennington, C., Picozzi, M., Ruhl, C.J., Shearer, P., Spallarossa, D., Trugman, D., Vandevert, I., Wu, Q., Yoon, C., Yu, E., Beroza, G.C., Eulenfeld, T., Knudson, T., Mayeda, K., Morasca, P., Neely, J.S., Roman-Nieves, J., Satriano, C., Supino, M., Walter, W.R., Archuleta, R., Atkinson, G.M., Calderoni, G., Ji, C., Yang, H., Zhang, J., 2025. Overview of the SCEC/USGS community stress drop validation study using the 2019 ridgecrest earthquake sequence. Bull. Seismol. Soc. Am. 115 (3), 734–759. https://doi.org/10.1785/0120240158

Achtziger-Zupančič, P., Ceccato, A., Zappone, A.S., Pozzi, G., Shakas, A., Amann, F., Behr, W.M., Escallon Botero, D., Giardini, D., Hertrich, M., Jalali, M., Ma, X., Meier, M.-A., Osten, J., Wiemer, S., Cocco, M., 2024. Selection and characterization of the target fault for fluid-induced activation and earthquake rupture experiments. Solid Earth 15 (8), 1087–1112. https://doi.org/10.5194/se-15-1087-2024

Aki, K., 1967. Scaling law of seismic spectrum. J. Geophys. Res. 72 (4), 1217–1231.

Aki, K., Richards, P.G., 2002. Quantitative Seismology. 2nd ed. University Science Books, Sausalito, CA.

Allmann, B.P., Shearer, P.M., 2009. Global variations of stress drop for moderate to large earthquakes. J. Geophys. Res. Solid Earth 114 (B1), B01310.

Andrews, D.J., 1976. Rupture velocity of plane strain shear cracks. J. Geophys. Res. 81 (32), 5679–5687.

Aochi, H., Madariaga, R., 2003. The 1999 Izmit, Turkey, earthquake: nonplanar fault structure, dynamic rupture process, and strong ground motion. Bull. Seismol. Soc. Am. 93 (3), 1249–1266.

Boore, D.M., Boatwright, J., 1984. Average body-wave radiation coefficients. Bull. Seismol. Soc. Am. 74 (5), 1615–1621.

Brune, J.N., 1970. Tectonic stress and the spectra of seismic shear waves from earthquakes. J. Geophys. Res. 75 (26), 4997–5009.

Chiaraluce, L., Amato, A., Carannante, S., Castelli, V., Cattaneo, M., Cocco, M., Collettini, C., D'Alema, E., Di Stefano, R., Latorre, D., et al., 2014. The alto tiberina near fault observatory (Northern apennines, Italy). Ann. Geophys. 57 (3), S0327.

Cocco, M., Aretusini, S., Cornelio, C., Nielsen, S.B., Spagnuolo, E., Tinti, E., Di Toro, G., 2023. Fracture energy and breakdown work during earthquakes. Annu. Rev. Earth Planet. Sci. 51 (1), 217–252.

Cocco, M., Tinti, E., Cirella, A., 2016. On the scale dependence of earthquake stress drop. J. Seismolog. 20, 1151–1170.

Dahlen, F.A., 1974. On the ratio of p-wave to s-wave corner frequencies for shallow earthquake sources. Bull. Seismol. Soc. Am. 64 (4), 1159–1180.

Das, S., Aki, K., 1977. Fault plane with barriers: a versatile earthquake model. J. Geophys. Res. 82 (36), 5658–5670.

Day, S.M., 1982. Three-dimensional simulation of spontaneous rupture: the effect of nonuniform prestress. Bull. Seismol. Soc. Am. 72 (6A), 1881–1902.

Deichmann, N., 2017. Theoretical basis for the observed break in ML/m w scaling between small and large earthquakes. Bull. Seismol. Soc. Am. 107 (2), 505–520.

Dreger, D., Nadeau, R.M., Chung, A., 2007. Repeating earthquake finite source models: strong asperities revealed on the san andreas fault. Geophys. Res. Lett. 34 (23), L23302.

Ellsworth, W.L., 2013. Injection-induced earthquakes. Science 341 (6142), 1225942.

Eshelby, J.D., 1957. The determination of the elastic field of an ellipsoidal inclusion, and related problems. Proc. R. Soc. London. Ser. A. Math. Phys. Sci. 241 (1226), 376–396.

Gabriel, A.-A., Garagash, D.I., Palgunadi, K.H., Mai, P.M., 2024. Fault size–dependent fracture energy explains multiscale seismicity and cascading earthquakes. Science 385 (6707), eadj9587.

Gabriel, A.-A., Kurapati, V., Niu, Z., Schliwa, N., Schneller, D., Ulrich, T., Dorozhinskii, R., Krenz, L., Uphoff, C., Wolf, S., Breuer, A., Heinecke, A., Pelties, C., Rettenberger, S., Wollherr, S., Bader, M., Seissol, 2025. https://doi.org/10.5281/zenodo.15685917.

Galis, M., Ampuero, J.P., Mai, P.M., Cappa, F., 2017. Induced seismicity provides insight into why earthquake ruptures stop. Sci. Adv. 3 (12), eaap7528.

Garagash, D.I., Germanovich, L.N., 2012. Nucleation and arrest of dynamic slip on a pressurized fault. J. Geophys. Res. Solid Earth 117 (B10), B10310.

Godano, M., Bernard, P., Dublanchet, P., 2015. Bayesian inversion of seismic spectral ratio for source scaling: application to a persistent multiplet in the western corinth rift. J. Geophys. Res. Solid Earth 120 (11), 7683–7712.

Guglielmi, Y., Cappa, F., Avouac, J.-P., Henry, P., Elsworth, D., 2015. Seismicity triggered by fluid injection–induced aseismic slip. Science 348 (6240), 1224–1226.

Hanks, T.C., Wyss, M., 1972. The use of body-wave spectra in the determination of seismic-source parameters. Bull. Seismol. Soc. Am. 62 (2), 561–589.

Ida, Y., 1972. Cohesive force across the tip of a longitudinal-shear crack and griffith's specific surface energy. J. Geophys. Res. 77 (20), 3796–3805.

Kame, N., Yamashita, T., 1999. A new light on arresting mechanism of dynamic earthquake faulting. Geophys. Res. Lett. 26 (13), 1997–2000.

Kanamori, H., 1977. The energy release in great earthquakes. J. Geophys. Res. 82 (20), 2981–2987.

Kanamori, H., Brodsky, E.E., 2004. The physics of earthquakes. Rep. Prog. Phys. 67 (8), 1429.

Kaneko, Y., Shearer, P.M., 2014. Seismic source spectra and estimated stress drop derived from cohesive-zone models of circular subshear rupture. Geophys. J. Int. 197 (2), 1002–1015.

Kaneko, Y., Shearer, P.M., 2015. Variability of seismic source spectra, estimated stress drop, and radiated energy, derived from cohesive-zone models of symmetrical and asymmetrical circular and elliptical ruptures. J. Geophys. Res. Solid Earth 120 (2), 1053–1079.

Ke, C.-Y., McLaskey, G.C., Kammer, D.S., 2022. Earthquake breakdown energy scaling despite constant fracture energy. Nat. Commun. 13 (1), 1005.

Lengliné, O., Lamourette, L., Vivin, L., Cuenot, N., Schmittbuhl, J., 2014. Fluid-induced earthquakes with variable stress drop. J. Geophys. Res. Solid Earth 119 (12), 8900–8913.

Lin, Y.-Y., Schaal, N., Lapusta, N., 2025. On seismological stress-drop estimates for earthquake sources simulated on rate-and-state faults. Bull. Seismol. Soc. Am. 20, 1–33.

Madariaga, R., 1976. Dynamics of an expanding circular fault. Bull. Seismol. Soc. Am. 66 (3), 639–666.

Madariaga, R., 1979. On the relation between seismic moment and stress drop in the presence of stress and strength heterogeneity. J. Geophys. Res. Solid Earth 84 (B5), 2243–2250.

Madariaga, R., Schubert, G., 2015. 4.02-Seismic source theory. Treatise Geophys. 4 (02), 51–71.

Malagnini, L., Mayeda, K., Nielsen, S., Yoo, S.-H., Munafo', I., Rawles, C., Boschi, E., 2014. Scaling transition in earthquake sources: a possible link between seismic and laboratory measurements. Pure Appl. Geophys. 171, 2685–2707.

Maurer, J., Dunham, E.M., Segall, P., 2020. Role of fluid injection on earthquake size in dynamic rupture simulations on rough faults. Geophys. Res. Lett. 47 (13), e2020GL088377.

McGarr, A., 2014. Maximum magnitude earthquakes induced by fluid injection. J. Geophys. Res. Solid Earth 119 (2), 1008–1019.

Mesimeri, M., Scarabello, L., Zimmermann, E., Haag, T., Zylis, E., Villiger, L., Kaestli, P., Meier, M.-A., Rinaldi, A.P., Obermann, A., et al., 2025. Multiscale seismic monitoring in the bedretto underground laboratory for geosciences and geoenergies (bulgg). Seismol. Res. Lett. 96 (1), 182–191.

Molnar, P., Tucker, B.E., Brune, J.N., 1973. Corner frequencies of p and s waves and models of earthquake sources. Bull. Seismol. Soc. Am. 63 (6-1), 2091–2104.
Mosconi, F., Tinti, E., Casarotti, E., Gabriel, A.-A., Rinaldi, A.P., Dal Zilio, L., Cocco, M., 2025. Modeling the 3D dynamic rupture of microearthquakes induced by fluid injection. J. Geophys. Res.. Solid Earth 130 (3), e2024JB029621.
Noda, H., Lapusta, N., Kanamori, H., 2013. Comparison of average stress drop measures for ruptures with heterogeneous stress change and implications for earthquake physics. Geophys J. Int. 193 (3), 1691–1712.
Norbeck, J.H., Horne, R.N., 2018. Maximum magnitude of injection-induced earthquakes: a criterion to assess the influence of pressure migration along faults. Tectonophysics 733, 108–118.
Oates, S., Schmitz, J., Zurek, B., Piesold, T., van Dedem, E., 2024. Empirical green's function analysis of some induced earthquake pairs from the groningen gas field. J. Seismolog. 28 (3), 579–613.
Obermann, A., Rosskopf, M., Durand, V., Plenkers, K., Bröker, K., Rinaldi, A.P., Gholizadeh Doonechaly, N., Gischig, V., Zappone, A., Amann, F., et al., 2024. Seismic response of hectometer-scale fracture systems to hydraulic stimulation in the bedretto underground laboratory, Switzerland. J. Geophys. Res. Solid Earth 129 (11), e2024JB029836.
Plenkers, K., Reinicke, A., Obermann, A., Gholizadeh Doonechaly, N., Krietsch, H., Fechner, T., Hertrich, M., Kontar, K., Maurer, H., Philipp, J., et al., 2023. Multidisciplinary monitoring networks for mesoscale underground experiments: advances in the bedretto reservoir project. Sensors 23 (6), 3315.
Poggiali, G., Chiaraluce, L., Ross, Z.E., Zhu, W., Marone, C., 2025. Fault geometry and source mechanics of the altotiberina fault system from a high-resolution machine-learning earthquake catalog. Bull. Seismol. Soc. Am. 115 (5), 2181–2201.
Prieto, G.A., Shearer, P.M., Vernon, F.L., Kilb, D., 2004. Earthquake source scaling and self-similarity estimation from stacking p and s spectra. J. Geophys. Res. Solid Earth 109 (B8), B08310.
Pulido, N., Dalguer, L.A., 2009. Estimation of the high-frequency radiation of the 2000 Tottori (Japan) earthquake based on a dynamic model of fault rupture: application to the strong ground motion simulation. Bull. Seismol. Soc. Am. 99 (4), 2305–2322.
Rinaldi, A.P., Rutqvist, J., Luu, K., Blanco-Martín, L., Hu, M., Sentís, M.L., Eberle, L., Kaestli, P., 2022. Tough3-flac3D: a modeling approach for parallel computing of fluid flow and geomechanics. Comput. Geosci. 26 (6), 1563–1580.
Ripperger, J., Ampuero, J.-P., Mai, P.M., Giardini, D., 2007. Earthquake source characteristics from dynamic rupture with constrained stochastic fault stress. J. Geophys. Res. Solid Earth 112 (B4), B04311.
Sáez, A., Lecampion, B., 2024. Fluid-driven slow slip and earthquake nucleation on a slip-weakening circular fault. J. Mech. Phys. Solids 183, 105506.
Sato, T., Hirasawa, T., 1973. Body wave spectra from propagating shear cracks. J. Phys. Earth 21 (4), 415–431.
Schliwa, N., Gabriel, A., 2023. Equivalent near-field corner frequency analysis of 3D dynamic rupture simulations reveals dynamic source effects. Seismol. Res. Lett. 95 (2A), 900–924. https://doi.org/10.1785/0220230225
Shearer, P.M., Vandevert, I., Fan, W., Abercrombie, R.E., Bindi, D., Calderoni, G., Chen, X., Ellsworth, W., Harrington, R., Huang, Y., Knudson, T., Roßbach, M., Satriano, C., Supino, M., Trugman, D.T., Yang, H., Zhang, J., 2024. Earthquake source spectra estimates vary widely for two ridgecrest aftershocks because of differences in attenuation corrections. Bull. Seismol. Soc. Am. 115 (3), 777–791. https://doi.org/10.1785/0120240134
Supino, M., Festa, G., Zollo, A., 2019. A probabilistic method for the estimation of earthquake source parameters from spectral inversion: application to the 2016–2017 Central Italy seismic sequence. Geophys. J. Int. 218 (2), 988–1007.
Supino, M., Poiata, N., Festa, G., Vilotte, J.-P., Satriano, C., Obara, K., 2020. Self-similarity of low-frequency earthquakes. Sci. Rep. 10 (1), 6523.
Supino, M., Scognamiglio, L., Chiaraluce, L., Doglioni, C., Herrero, A., 2024. Source characterization of the 20th May 2024 MD 4.4 campi flegrei caldera earthquake through a joint source-propagation probabilistic inversion. Seismica 3 (2).
Uchida, N., Matsuzawa, T., Ellsworth, W.L., Imanishi, K., Okada, T., Hasegawa, A., 2007. Source parameters of a m4. 8 and its accompanying repeating earthquakes off Kamaishi, NE Japan: implications for the hierarchical structure of asperities and earthquake cycle. Geophys. Res. Lett. 34 (20), L20313.
Udías, A., Madariaga, R., Buforn, E., 2014. Source Mechanisms of Earthquakes: Theory and Practice. Cambridge University Press.
Volpe, G., Pozzi, G., Collettini, C., Spagnuolo, E., Achtziger-Zupančič, P., Zappone, A., Aldega, L., Meier, M.-A., Giardini, D., Cocco, M., 2023. Laboratory simulation of fault reactivation by fluid injection and implications for induced seismicity at the bedretto-lab, Swiss Alps. Tectonophysics 862, 229987.
Wang, Y., Day, S.M., 2017. Seismic source spectral properties of crack-like and pulse-like modes of dynamic rupture. J. Geophys. Res. Solid Earth 122 (8), 6657–6684.
Xu, D., Gong, W., Zhang, Z., Xu, J., Yu, H., Chen, X., 2023. The 2016 menyuan earthquake: the largest self-arrested crustal earthquake ever observed. Geophys. Res. Lett. 50 (11), e2023GL103556.
Zoback, M., Hickman, S., Ellsworth, W., S. S. Team, et al., 2011. Scientific drilling into the san andreas fault zone–an overview of SAFOD's first five years. Sci. Drill. 11, 14–28.